

Vivriti Co-Lending Control Tower: Clean Environment Runbook & Verification Guide

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1. Prerequisites & Environment Setup

The application has been verified from a completely clean environment on both **Windows (PowerShell / Command Prompt)** and **Linux / macOS (Bash)**.

1.1 Minimum System Requirements

- **Java Development Kit (JDK):** Version **21** (Temurin, Oracle, Corretto, or Microsoft Build of OpenJDK).
- **Build Tool:** Bundled Maven Wrapper (`./mvnw` on Linux/macOS or `.\mvnw.cmd` on Windows) — no manual Maven installation required.
- **Database:**
- **Production Profile (`default`):** PostgreSQL 16 (listening on `localhost:5432` with user `postgres` , database `controlltower`).
- **Testing Profile (`test`):** In-Memory H2 database (automatically spun up with zero external dependencies).
- **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox for the interactive operations dashboard.

2. One-Click Execution Scripts

2.1 Windows Clean Run

Open a Command Prompt or PowerShell terminal in the repository root directory:

```
run.bat
```

The script automatically detects Java 21, checks PostgreSQL availability (falling back to H2 if offline), applies schema migrations, generates 1,000 synthetic loans, executes ingestion and 5-tier reconciliation, runs gate evaluation, and executes the 13 automated tests.

2.2 Linux / macOS Clean Run

Open a terminal in the repository root directory:

```
chmod +x run.sh scripts/*.sh
./run.sh
```

3. Manual Step-by-Step Execution

Step 1: Run the Automated Hard-Gate Test Suite

Execute the entire test suite covering failure isolation, determinism, idempotency, close equation breaks, and maker-checker segregation:

```
# Windows
.\mvnw.cmd test

# Linux / macOS
./mvnw test
```

Expected Output:

```
[INFO] Results:
[INFO] Tests run: 13, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
```

Step 2: Start the Control Tower Application

Launch the Spring Boot application on port 8080:

```
# Windows
.\mvnw.cmd spring-boot:run

# Linux / macOS
./mvnw spring-boot:run
```

Wait until the console logs show:

```
Started ControlTowerApplication in X.XXX seconds
Tomcat started on port 8080 (http) with context path '/'
```

Step 3: Access the Operations Dashboard

Open your browser and navigate to:

```
http://localhost:8080
```

(No login required for dashboard landing; user profile switcher controls role-based actions).

4. Interactive Web UI Walkthrough & Verification

4.1 Workstream 1: Synthetic Data Generation & Data Quality Report

- On the left sidebar or in the Operations Center, click **1. Generate Feeds**.
- To inspect the generated data quality report:
- Click the new " **Data Quality Report**" item on the left sidebar.
- Or click " **View Quality Report**" in the Operations Center.
- Review the live metrics:
- **Total Loans:** 2,000 unique loans.
- **Feed Records:** 5,972 multi-format records across Partner Alpha (JSON), Beta (CSV), Gamma (Pipe).
- **Overall Anomaly Rate:** $\geq 5.00\%$ (e.g. 9.40%).
- **All 10 Anomaly Classes:** Missing Bank Leg, Missing LMS Leg, Duplicate Callback, Amount Mismatch, Status Mismatch, Timing Difference, Composite Split, Orphan Reversal, Schema Breach, Batch Total Drift.
- **Raw Markdown:** View line-by-line or click " **Copy Markdown**".

4.2 Workstream 2: Multi-Format Feed Ingestion

- In the Operations Center, click **2. Ingest Multi-Format**.
- Inspect the terminal stream:
- SHA-256 payload hashes computed for all records.
- Idempotency guard validates zero duplicates.
- Schema breaches isolated to `raw_source_record` with `quarantined=true`.
- Valid records normalized into `canonical_event`.

4.3 Workstream 3: 5-Tier Reconciliation Engine

- Click **3. 5-Tier Reconcile**.
- Engine processes events through the chain of responsibility:
- Exact 3-leg matches reconciled straight-through.
- Composite splits balanced across multiple settlement debits.
- Post-cutoff callbacks placed in 24h grace window (`PENDING_GRACE_PERIOD`).
- Unresolved breaks routed to the Exception Queue.

4.4 Workstream 4: Gate Scorecard & Evaluation

- Click **4. Evaluate Gate** or navigate to " **Gate Scorecard**".
- Verify the hard gate:
- **False Matches: 0 Loans** (Tolerance: 0).
- **False Match Financial Exposure: ₹0.00** (Tolerance: ₹0.00).
- **Gate Status: PASSED** (in green).
- Inspect the Confusion Matrix across all 10 anomaly types.

4.5 Workstream 5: Batch Close & Maker-Checker Verification

- Navigate to " **Batch Close Workbench**".
- Select batch `B-PARTNER_ALPHA-BANK-20240115` from the dropdown:
- Click **Evaluate Close**.

- Notice the decision renders: **DECISION: HOLD** (in bold red).
- Reason: The batch contains quarantined records and partner exceptions. The equation identifies all blocking exceptions and halts close.
- Switch the dropdown to a clean validated batch:
- Click **Evaluate Close**.
- Decision renders: **DECISION: CLOSE** (in green) with ₹0.00 unaccounted delta.
- Test Four-Eyes Segregation of Duties:
- Switch active profile to **Operator (Maker)** using the top-right profile avatar.
- Attempt to click **Approve Close** → blocked with HTTP 403.
- Switch to **Approver (Checker)** → approval succeeds only if this user did not execute an override on the same batch.

4.6 Workstream 6: Audit Lineage Trace

- Navigate to "📄 **Audit Lineage Trace**".
- Enter loan reference **LOAN-000001** and click **Trace Lineage**.
- View the complete cryptographic chain:

Raw Source Payload → SHA-256 Hash → Canonical Event → Reconciliation Decision

5. Production Observability Endpoints

All actuator and documentation endpoints are exposed and operational:

Endpoint	URL	Description
System Logfile	http://localhost:8080/actuator/logfile	Live SLF4J application log stream including evaluation scorecards.
System Health	http://localhost:8080/actuator/health	PostgreSQL connectivity, disk space, and application liveness.
Swagger / OpenAPI	http://localhost:8080/swagger-ui/index.html	Interactive REST API testing console with OpenAPI 3.0 schema.
Dynamic Loggers	http://localhost:8080/actuator/loggers	Real-time adjustment of log levels (TRACE, DEBUG, INFO) without restart.
JVM Metrics	http://localhost:8080/actuator/metrics	Detailed metrics for memory, threads, garbage collection, and HikariCP.
HTTP Trace	http://localhost:8080/actuator/httpexchanges	History of recent HTTP request/response exchanges.

6. Verification Checklist Summary

- [x] Java 21 compile clean with zero compilation warnings or errors.
 - [x] All 13 unit, architecture, and integration tests pass cleanly with `BUILD SUCCESS`.
 - [x] ArchUnit test enforces strict isolation of Ground Truth from reconciliation engine.
 - [x] Batch Close evaluates to `HOLD` on quarantined batches and `CLOSE` on balanced batches.
 - [x] Quality Report generated and viewable in UI on every run.
 - [x] Four-Eyes governance strictly prevents Maker from acting as Checker.
 - [x] Zero false matches, zero false exposure, and zero unaccounted paise verified.
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Vivriti Co-Lending Control Tower Operations Runbook — Verified Clean Environment.